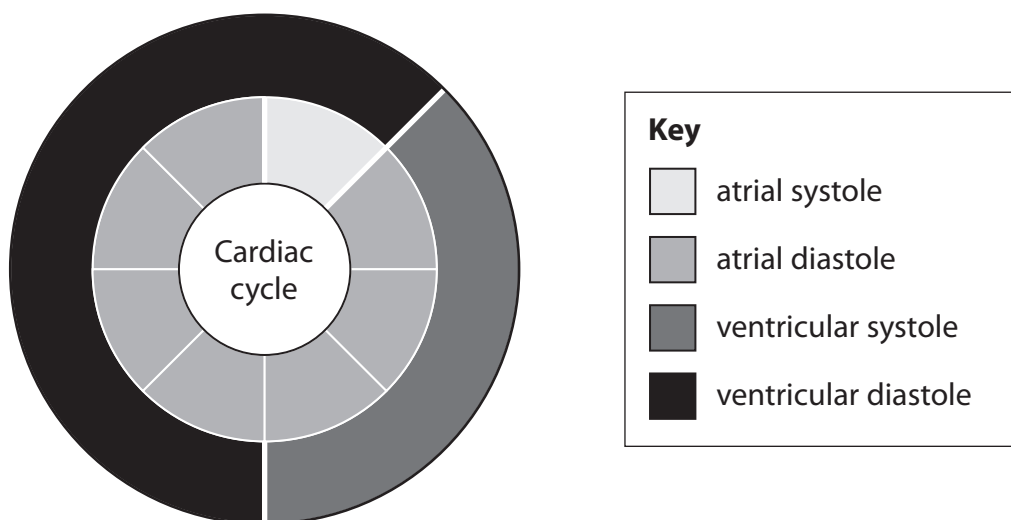


3 The cardiac cycle is the sequence of events that occurs when the heart beats.

A typical cardiac cycle takes 0.86 seconds.

(a) The diagram illustrates the cardiac cycle.



(i) Which row of the table describes the atria and ventricles during atrial systole?

(1)

|                                   | Atria      | Ventricles |
|-----------------------------------|------------|------------|
| <input type="checkbox"/> <b>A</b> | contracted | contracted |
| <input type="checkbox"/> <b>B</b> | contracted | relaxed    |
| <input type="checkbox"/> <b>C</b> | relaxed    | contracted |
| <input type="checkbox"/> <b>D</b> | relaxed    | relaxed    |

(ii) Explain why there is a delay of 0.01 seconds between atrial systole and ventricular systole.

(2)

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



- (iii) Using the information in the diagram, calculate the duration of ventricular systole in milliseconds.

Express your answer in standard form.

(2)

Answer ..... ms

- (iv) State what proportion of the cardiac cycle is spent in ventricular diastole.

(1)

- (b) A typical cardiac cycle takes 0.86 seconds.

During exercise, the heart rate increases and the duration of the cardiac cycle decreases.

Calculate the increase in heart rate if the cardiac cycle decreases by 0.4 seconds.

(3)

Answer ..... beats per minute

**(Total for Question 3 = 9 marks)**

